2.06 Complex Numbers: The nth Roots of a Complex Number

Worked Solutions

1 a)

$$z^3 = 8\mathrm{i}$$

Using modulus-argument form:

$$8\mathrm{i} = 8\left(\cos\frac{\pi}{2} + \mathrm{i}\sin\frac{\pi}{2}\right)$$

$$z^3 = r^3(\cos\theta + \mathrm{i}\sin\theta)^3 = r^3(\cos 3\theta + \mathrm{i}\sin 3\theta)$$

$$r^3 = 8 \Rightarrow r = 2$$

Since $3\theta = \frac{\pi}{2} + 2k\pi$ for integer k:

$$\theta = \frac{\pi}{6} + \frac{2k\pi}{3}$$

$$\begin{aligned} k = 0: &\quad z_0 = 2\mathrm{e}^{\pi\mathrm{i}/6} = \sqrt{3} + \mathrm{i} \\ k = 1: &\quad z_1 = 2\mathrm{e}^{5\pi\mathrm{i}/6} = -\sqrt{3} + \mathrm{i} \\ k = 2: &\quad z_2 = 2\mathrm{e}^{3\pi\mathrm{i}/2} = 2\mathrm{e}^{-\pi\mathrm{i}/2} = -2\mathrm{i} \end{aligned}$$

$$\therefore z = \sqrt{3} + \mathrm{i}, \quad z = -\sqrt{3} + \mathrm{i} \quad \text{and} \quad z = -2\mathrm{i}.$$

1 b)

$$z^3 = -27$$

Using modulus-argument form:

$$-27 = 27(\cos\pi + \mathrm{i}\sin\pi)$$

$$z^3 = r^3(\cos\theta + \mathrm{i}\sin\theta)^3 = r^3(\cos 3\theta + \mathrm{i}\sin 3\theta)$$

$$r^3 = 27 \Rightarrow r = 3$$

Since $3\theta = \pi + 2k\pi$ for integer k:

$$\theta = \frac{\pi}{3} + \frac{2k\pi}{3}$$

$$\begin{aligned} k = 0: &\quad z_0 = 3\mathrm{e}^{\pi\mathrm{i}/3} = \frac{3}{2} + \frac{3\sqrt{3}}{2}\mathrm{i} \\ k = 1: &\quad z_1 = 3\mathrm{e}^{\pi\mathrm{i}} = -3 \\ k = 2: &\quad z_2 = 3\mathrm{e}^{5\pi\mathrm{i}/3} = 3\mathrm{e}^{-\pi\mathrm{i}/3} = \frac{3}{2} - \frac{3\sqrt{3}}{2}\mathrm{i} \end{aligned}$$

$$\therefore z = \frac{3}{2} + \frac{3\sqrt{3}}{2}\mathrm{i}, \quad z = -3 \quad \text{and} \quad z = \frac{3}{2} - \frac{3\sqrt{3}}{2}\mathrm{i}.$$

1 c)

$$z^2 = 1 + \sqrt{3}\mathrm{i}$$

Using modulus-argument form:

$$1 + \sqrt{3}\mathrm{i} = 2\left(\cos\frac{\pi}{3} + \mathrm{i}\sin\frac{\pi}{3}\right)$$

$$z^2 = r^2(\cos\theta + \mathrm{i}\sin\theta)^2 = r^2(\cos 2\theta + \mathrm{i}\sin 2\theta)$$

$$r^2 = 2 \Rightarrow r = \sqrt{2}$$

Since $2\theta = \frac{\pi}{3} + 2k\pi$ for integer k:

$$\theta = \frac{\pi}{6} + \frac{2k\pi}{2}$$

$$k = 0: \quad z_0 = \sqrt{2}\mathrm{e}^{\pi\mathrm{i}/6} = \frac{\sqrt{6}}{2} + \frac{\sqrt{2}}{2}\mathrm{i}$$

$$k = 1: \quad z_1 = \sqrt{2}\mathrm{e}^{7\pi\mathrm{i}/6} = \sqrt{2}\mathrm{e}^{-5\pi\mathrm{i}/6} = -\frac{\sqrt{6}}{2} - \frac{\sqrt{2}}{2}\mathrm{i}$$

$$\therefore z = \frac{\sqrt{6}}{2} + \frac{\sqrt{2}}{2}\mathrm{i} \quad \text{and} \quad z = -\frac{\sqrt{6}}{2} - \frac{\sqrt{2}}{2}\mathrm{i}.$$

1 d)

$$z^4 = -16$$

Using modulus-argument form:

$$-16 = 16(\cos(-\pi) + \mathrm{i}\sin(-\pi))$$

$$z^4 = r^4(\cos\theta + \mathrm{i}\sin\theta)^4 = r^4(\cos 4\theta + \mathrm{i}\sin 4\theta)$$

$$r^4 = 16 \Rightarrow r = 2$$

Since $4\theta = -\pi + 2k\pi$ for integer k:

$$\theta = -\frac{\pi}{4} + \frac{2k\pi}{4}$$

$$k = 0: \quad z_0 = 2\mathrm{e}^{-\pi\mathrm{i}/4} = \sqrt{2} - \sqrt{2}\mathrm{i}$$
$$k = 1: \quad z_1 = 2\mathrm{e}^{\pi\mathrm{i}/4} = \sqrt{2} + \sqrt{2}\mathrm{i}$$
$$k = 2: \quad z_2 = 2\mathrm{e}^{3\pi\mathrm{i}/4} = -\sqrt{2} + \sqrt{2}\mathrm{i}$$
$$k = 3: \quad z_3 = 2\mathrm{e}^{5\pi\mathrm{i}/4} = 2\mathrm{e}^{-3\pi\mathrm{i}/4} = -\sqrt{2} - \sqrt{2}\mathrm{i}$$

$$\therefore z = \sqrt{2} - \sqrt{2}\mathrm{i}, \quad z = \sqrt{2} + \sqrt{2}\mathrm{i}, \quad z = -\sqrt{2} + \sqrt{2}\mathrm{i} \quad \text{and} \quad z = -\sqrt{2} - \sqrt{2}\mathrm{i}.$$

2 a)

$$z^5 = 32\mathrm{i}$$

Using modulus-argument form:

$$32\mathrm{i} = 32\left(\cos\frac{\pi}{2} + \mathrm{i}\sin\frac{\pi}{2}\right)$$

$$z^5 = r^5(\cos\theta + \mathrm{i}\sin\theta)^5 = r^5(\cos 5\theta + \mathrm{i}\sin 5\theta)$$

$$r^5 = 32 \Rightarrow r = 2$$

Since $5\theta = \frac{\pi}{2} + 2k\pi$ for integer k:

$$\theta = \frac{\pi}{10} + \frac{2k\pi}{5}$$

$$\begin{aligned}
&k = 0 : z_0 = 2(\cos\tfrac{\pi}{10} + \mathrm{i}\sin\tfrac{\pi}{10}) \\
&k = 1 : z_1 = 2(\cos\tfrac{\pi}{2} + \mathrm{i}\sin\tfrac{\pi}{2}) \\
&k = 2 : z_2 = 2(\cos\tfrac{9\pi}{10} + \mathrm{i}\sin\tfrac{9\pi}{10}) \\
&k = 3 : z_3 = 2(\cos\tfrac{13\pi}{10} + \mathrm{i}\sin\tfrac{13\pi}{10}) = 2(\cos(-\tfrac{7\pi}{10}) + \mathrm{i}\sin(-\tfrac{7\pi}{10})) \\
&k = 4 : z_4 = 2(\cos\tfrac{17\pi}{10} + \mathrm{i}\sin\tfrac{17\pi}{10}) = 2(\cos(-\tfrac{3\pi}{10}) + \mathrm{i}\sin(-\tfrac{3\pi}{10}))
\end{aligned}$$

$$\therefore z = 2(\cos\tfrac{\pi}{10} + \mathrm{i}\sin\tfrac{\pi}{10}),\ z = 2(\cos\tfrac{\pi}{2} + \mathrm{i}\sin\tfrac{\pi}{2}),\ z = 2(\cos\tfrac{9\pi}{10} + \mathrm{i}\sin\tfrac{9\pi}{10}),$$
$$z = 2(\cos(-\tfrac{7\pi}{10}) + \mathrm{i}\sin(-\tfrac{7\pi}{10})) \quad \text{and} \quad z = 2(\cos(-\tfrac{3\pi}{10}) + \mathrm{i}\sin(-\tfrac{3\pi}{10})).$$

2 b)

$$z^3 = 1 + \sqrt{3}\mathrm{i}$$

Using modulus-argument form:

$$1 + \sqrt{3}\mathrm{i} = 2(\cos\tfrac{\pi}{3} + \mathrm{i}\sin\tfrac{\pi}{3})$$

$$z^3 = r^3(\cos\theta + \mathrm{i}\sin\theta)^3 = r^3(\cos 3\theta + \mathrm{i}\sin 3\theta)$$

$$r^3 = 2 \Rightarrow r = 2^{1/3}$$

Since $3\theta = \frac{\pi}{3} + 2k\pi$ for integer k:

$$\theta = \frac{\pi}{9} + \frac{2k\pi}{3}$$

$$\begin{aligned}
&k = 0 : z_0 = 2^{1/3}(\cos\tfrac{\pi}{9} + \mathrm{i}\sin\tfrac{\pi}{9}) \\
&k = 1 : z_1 = 2^{1/3}(\cos\tfrac{7\pi}{9} + \mathrm{i}\sin\tfrac{7\pi}{9}) \\
&k = 2 : z_2 = 2^{1/3}(\cos\tfrac{13\pi}{9} + \mathrm{i}\sin\tfrac{13\pi}{9}) = 2^{1/3}(\cos(-\tfrac{5\pi}{9}) + \mathrm{i}\sin(-\tfrac{5\pi}{9}))
\end{aligned}$$

$$\therefore z = 2^{1/3}(\cos\tfrac{\pi}{9} + \mathrm{i}\sin\tfrac{\pi}{9}), \quad z = 2^{1/3}(\cos\tfrac{7\pi}{9} + \mathrm{i}\sin\tfrac{7\pi}{9})$$
$$\text{and } z = 2^{1/3}(\cos(-\tfrac{5\pi}{9}) + \mathrm{i}\sin(-\tfrac{5\pi}{9})).$$

2 c)

$$z^4 = -2\sqrt{2} + 2\sqrt{2}\mathrm{i}$$

Using modulus-argument form:

$$-2\sqrt{2} + 2\sqrt{2}\mathrm{i} = 4(\cos\tfrac{3\pi}{4} + \mathrm{i}\sin\tfrac{3\pi}{4})$$
$$z^4 = r^4(\cos\theta + \mathrm{i}\sin\theta)^4 = r^4(\cos 4\theta + \mathrm{i}\sin 4\theta)$$
$$r^4 = 4 \Rightarrow r = 4^{1/4} = \sqrt{2}$$

Since $4\theta = \frac{3\pi}{4} + 2k\pi$ for integer k:

$$\theta = \frac{3\pi}{16} + \frac{2k\pi}{4}$$

$$k = 0 : z_0 = \sqrt{2}(\cos\tfrac{3\pi}{16} + \mathrm{i}\sin\tfrac{3\pi}{16})$$
$$k = 1 : z_1 = \sqrt{2}(\cos\tfrac{11\pi}{16} + \mathrm{i}\sin\tfrac{11\pi}{16})$$
$$k = 2 : z_2 = \sqrt{2}(\cos\tfrac{19\pi}{16} + \mathrm{i}\sin\tfrac{19\pi}{16}) = \sqrt{2}(\cos(-\tfrac{13\pi}{16}) + \mathrm{i}\sin(-\tfrac{13\pi}{16}))$$
$$k = 3 : z_3 = \sqrt{2}(\cos\tfrac{27\pi}{16} + \mathrm{i}\sin\tfrac{27\pi}{16}) = \sqrt{2}(\cos(-\tfrac{5\pi}{16}) + \mathrm{i}\sin(-\tfrac{5\pi}{16}))$$

$$\therefore z = \sqrt{2}(\cos\tfrac{3\pi}{16} + \mathrm{i}\sin\tfrac{3\pi}{16}),\ z = \sqrt{2}(\cos\tfrac{11\pi}{16} + \mathrm{i}\sin\tfrac{11\pi}{16}),$$
$$z = \sqrt{2}(\cos(-\tfrac{13\pi}{16}) + \mathrm{i}\sin(-\tfrac{13\pi}{16})) \quad \text{and} \quad z = \sqrt{2}(\cos(-\tfrac{5\pi}{16}) + \mathrm{i}\sin(-\tfrac{5\pi}{16})).$$

2 d)

$$z^8 = 8 + 8\sqrt{3}\mathrm{i}$$

Using modulus-argument form:

$$8 + 8\sqrt{3}\mathrm{i} = 16(\cos\tfrac{\pi}{3} + \mathrm{i}\sin\tfrac{\pi}{3})$$
$$z^8 = r^8(\cos\theta + \mathrm{i}\sin\theta)^8 = r^8(\cos 8\theta + \mathrm{i}\sin 8\theta)$$
$$r^8 = 16 \Rightarrow r = 16^{1/8} = \sqrt{2}$$

Since $8\theta = \frac{\pi}{3} + 2k\pi$ for integer k:

$$\theta = \frac{\pi}{24} + \frac{2k\pi}{8}$$

$$k = 0 : z_0 = \sqrt{2}(\cos\tfrac{\pi}{24} + \mathrm{i}\sin\tfrac{\pi}{24})$$
$$k = 1 : z_1 = \sqrt{2}(\cos\tfrac{7\pi}{24} + \mathrm{i}\sin\tfrac{7\pi}{24})$$
$$k = 2 : z_2 = \sqrt{2}(\cos\tfrac{13\pi}{24} + \mathrm{i}\sin\tfrac{13\pi}{24})$$
$$k = 3 : z_3 = \sqrt{2}(\cos\tfrac{19\pi}{24} + \mathrm{i}\sin\tfrac{19\pi}{24})$$
$$k = 4 : z_4 = \sqrt{2}(\cos\tfrac{25\pi}{24} + \mathrm{i}\sin\tfrac{25\pi}{24}) = \sqrt{2}(\cos(-\tfrac{23\pi}{24}) + \mathrm{i}\sin(-\tfrac{23\pi}{24}))$$
$$k = 5 : z_5 = \sqrt{2}(\cos\tfrac{31\pi}{24} + \mathrm{i}\sin\tfrac{31\pi}{24}) = \sqrt{2}(\cos(-\tfrac{17\pi}{24}) + \mathrm{i}\sin(-\tfrac{17\pi}{24}))$$
$$k = 6 : z_6 = \sqrt{2}(\cos\tfrac{37\pi}{24} + \mathrm{i}\sin\tfrac{37\pi}{24}) = \sqrt{2}(\cos(-\tfrac{11\pi}{24}) + \mathrm{i}\sin(-\tfrac{11\pi}{24}))$$
$$k = 7 : z_7 = \sqrt{2}(\cos\tfrac{43\pi}{24} + \mathrm{i}\sin\tfrac{43\pi}{24}) = \sqrt{2}(\cos(-\tfrac{5\pi}{24}) + \mathrm{i}\sin(-\tfrac{5\pi}{24}))$$

$$\therefore z = \sqrt{2}(\cos\tfrac{\pi}{24} + \mathrm{i}\sin\tfrac{\pi}{24}),\ z = \sqrt{2}(\cos\tfrac{7\pi}{24} + \mathrm{i}\sin\tfrac{7\pi}{24}),$$
$$z = \sqrt{2}(\cos\tfrac{13\pi}{24} + \mathrm{i}\sin\tfrac{13\pi}{24}),\ z = \sqrt{2}(\cos\tfrac{19\pi}{24} + \mathrm{i}\sin\tfrac{19\pi}{24}),$$
$$z = \sqrt{2}(\cos(-\tfrac{23\pi}{24}) + \mathrm{i}\sin(-\tfrac{23\pi}{24})),\ z = \sqrt{2}(\cos(-\tfrac{17\pi}{24}) + \mathrm{i}\sin(-\tfrac{17\pi}{24})),$$
$$z = \sqrt{2}(\cos(-\tfrac{11\pi}{24}) + \mathrm{i}\sin(-\tfrac{11\pi}{24})) \quad \text{and} \quad z = \sqrt{2}(\cos(-\tfrac{5\pi}{24}) + \mathrm{i}\sin(-\tfrac{5\pi}{24})).$$

3 a)

$$z^7 = 128\mathrm{i}$$

Using modulus-argument form:

$$128\mathrm{i} = 128(\cos\tfrac{\pi}{2} + \mathrm{i}\sin\tfrac{\pi}{2})$$
$$z^7 = r^7(\cos\theta + \mathrm{i}\sin\theta)^7 = r^7(\cos 7\theta + \mathrm{i}\sin 7\theta)$$
$$r^7 = 128 \Rightarrow r = 2$$

Since $7\theta = \frac{\pi}{2} + 2k\pi$ for integer k:

$$\theta = \frac{\pi}{14} + \frac{2k\pi}{7}$$

$$\begin{aligned}
k = 0: &\quad z_0 = 2\mathrm{e}^{\pi\mathrm{i}/14} \\
k = 1: &\quad z_1 = 2\mathrm{e}^{5\pi\mathrm{i}/14} \\
k = 2: &\quad z_2 = 2\mathrm{e}^{9\pi\mathrm{i}/14} \\
k = 3: &\quad z_3 = 2\mathrm{e}^{13\pi\mathrm{i}/14} \\
k = 4: &\quad z_4 = 2\mathrm{e}^{17\pi\mathrm{i}/14} = 2\mathrm{e}^{-11\pi\mathrm{i}/14} \\
k = 5: &\quad z_5 = 2\mathrm{e}^{21\pi\mathrm{i}/14} = 2\mathrm{e}^{-\pi\mathrm{i}/2} \\
k = 6: &\quad z_6 = 2\mathrm{e}^{25\pi\mathrm{i}/14} = 2\mathrm{e}^{-3\pi\mathrm{i}/14}
\end{aligned}$$

$$\begin{aligned}
\therefore\ & z = 2\mathrm{e}^{\pi\mathrm{i}/14},\ z = 2\mathrm{e}^{5\pi\mathrm{i}/14},\ z = 2\mathrm{e}^{9\pi\mathrm{i}/14}, \\
& z = 2\mathrm{e}^{13\pi\mathrm{i}/14},\ z = 2\mathrm{e}^{-11\pi\mathrm{i}/14},\ z = 2\mathrm{e}^{-\pi\mathrm{i}/2},\ z = 2\mathrm{e}^{-3\pi\mathrm{i}/14}.
\end{aligned}$$

3 b)

$$z^6 = -32 - 32\sqrt{3}\mathrm{i}$$

Using modulus-argument form:

$$-32 - 32\sqrt{3}\mathrm{i} = 64\left(\cos(-\tfrac{2\pi}{3}) + \mathrm{i}\sin(-\tfrac{2\pi}{3})\right)$$
$$z^6 = r^6(\cos\theta + \mathrm{i}\sin\theta)^6 = r^6(\cos 6\theta + \mathrm{i}\sin 6\theta)$$
$$r^6 = 64 \Rightarrow r = 2$$

Since $6\theta = -\frac{2\pi}{3} + 2k\pi$ for integer k:

$$\theta = -\frac{2\pi}{18} + \frac{2k\pi}{6}$$

$$\begin{aligned}
k = 0: z_0 &= 2\mathrm{e}^{-\pi\mathrm{i}/9} \\
k = 1: z_1 &= 2\mathrm{e}^{2\pi\mathrm{i}/9} \\
k = 2: z_2 &= 2\mathrm{e}^{5\pi\mathrm{i}/9} \\
k = 3: z_3 &= 2\mathrm{e}^{8\pi\mathrm{i}/9} \\
k = 4: z_4 &= 2\mathrm{e}^{11\pi\mathrm{i}/9} = 2\mathrm{e}^{-7\pi\mathrm{i}/9} \\
k = 5: z_5 &= 2\mathrm{e}^{14\pi\mathrm{i}/9} = 2\mathrm{e}^{-4\pi\mathrm{i}/9}
\end{aligned}$$

$$\therefore\ z = 2\mathrm{e}^{-\pi\mathrm{i}/9},\ z = 2\mathrm{e}^{2\pi\mathrm{i}/9},\ z = 2\mathrm{e}^{5\pi\mathrm{i}/9},\ z = 2\mathrm{e}^{8\pi\mathrm{i}/9},\ z = 2\mathrm{e}^{-7\pi\mathrm{i}/9} \quad \text{and} \quad z = 2\mathrm{e}^{-4\pi\mathrm{i}/9}.$$

3 c)

$$z^4 = \frac{9}{2} + \frac{9}{2}\sqrt{3}\mathrm{i}$$

Using modulus-argument form:

$$\frac{9}{2} + \frac{9}{2}\sqrt{3}\mathrm{i} = 9(\cos\tfrac{\pi}{3} + \mathrm{i}\sin\tfrac{\pi}{3})$$

$$z^4 = r^4(\cos\theta + \mathrm{i}\sin\theta)^4 = r^4(\cos 4\theta + \mathrm{i}\sin 4\theta)$$

$$r^4 = 9 \Rightarrow r = \sqrt{3}$$

Since $4\theta = \frac{\pi}{3} + 2k\pi$ for integer k:

$$\theta = \frac{\pi}{12} + \frac{2k\pi}{4}$$

$$\begin{aligned} k = 0: &\quad z_0 = \sqrt{3}\mathrm{e}^{\pi\mathrm{i}/12} \\ k = 1: &\quad z_1 = \sqrt{3}\mathrm{e}^{7\pi\mathrm{i}/12} \\ k = 2: &\quad z_2 = \sqrt{3}\mathrm{e}^{13\pi\mathrm{i}/12} = \sqrt{3}\mathrm{e}^{-11\pi\mathrm{i}/12} \\ k = 3: &\quad z_3 = \sqrt{3}\mathrm{e}^{19\pi\mathrm{i}/12} = \sqrt{3}\mathrm{e}^{-5\pi\mathrm{i}/12} \end{aligned}$$

$$\therefore z = \sqrt{3}\mathrm{e}^{\pi\mathrm{i}/12},\ z = \sqrt{3}\mathrm{e}^{7\pi\mathrm{i}/12},\ z = \sqrt{3}\mathrm{e}^{-11\pi\mathrm{i}/12} \quad \text{and} \quad z = \sqrt{3}\mathrm{e}^{-5\pi\mathrm{i}/12}.$$

4)

$$z^3 = -2 + 2\mathrm{i}$$

Using modulus-argument form:

$$-2 + 2\mathrm{i} = 2\sqrt{2}(\cos\tfrac{3\pi}{4} + \mathrm{i}\sin\tfrac{3\pi}{4})$$

$$z^3 = r^3(\cos\theta + \mathrm{i}\sin\theta)^3 = r^3(\cos 3\theta + \mathrm{i}\sin 3\theta)$$

$$r^3 = 2\sqrt{2} \Rightarrow r = \sqrt{2}$$

Since $3\theta = \frac{3\pi}{4} + 2k\pi$ for integer k, $\theta = \frac{3\pi}{12} + \frac{2k\pi}{3}$:

$$\begin{aligned} k = 0: &\quad z_0 = \sqrt{2}\mathrm{e}^{\pi\mathrm{i}/4} = 1 + \mathrm{i} \\ k = 1: &\quad z_1 = \sqrt{2}\mathrm{e}^{11\pi\mathrm{i}/12} = -1.37 + 0.366\mathrm{i} \\ k = 2: &\quad z_2 = \sqrt{2}\mathrm{e}^{19\pi\mathrm{i}/12} = \sqrt{2}\mathrm{e}^{-5\pi\mathrm{i}/12} = 0.366 - 1.37\mathrm{i} \end{aligned}$$

$$\therefore z = 1 + \mathrm{i}, \quad z = -1.37 + 0.366\mathrm{i} \quad \text{and} \quad z = 0.366 - 1.37\mathrm{i}.$$

5 a)

$$z^2 = -2 + 2\mathrm{i}$$

Using modulus-argument form:

$$-2 + 2\mathrm{i} = 2\sqrt{2}(\cos\tfrac{3\pi}{4} + \mathrm{i}\sin\tfrac{3\pi}{4})$$

$$z^2 = r^2(\cos\theta + \mathrm{i}\sin\theta)^2 = r^2(\cos 2\theta + \mathrm{i}\sin 2\theta)$$

$$r^2 = 2\sqrt{2} \Rightarrow r = 2^{3/4}$$

Since $2\theta = \frac{3\pi}{4} + 2k\pi$ for integer k:

$$\theta = \frac{3\pi}{8} + \frac{2k\pi}{2}$$

$$k = 0: \quad z_0 = 2^{3/4}\mathrm{e}^{3\pi\mathrm{i}/8} = 0.644 + 1.55\mathrm{i}$$
$$k = 1: \quad z_1 = 2^{3/4}\mathrm{e}^{11\pi\mathrm{i}/8} = 2^{3/4}\mathrm{e}^{-5\pi\mathrm{i}/8} = -0.644 - 1.55\mathrm{i}$$

$$\therefore z = 0.644 + 1.55\mathrm{i} \quad \text{and} \quad z = -0.644 - 1.55\mathrm{i}.$$

5 b)

$$z^2 = -2 + 4\mathrm{i}$$

Using modulus-argument form:

$$-2 + 4\mathrm{i} = 2\sqrt{5}(\cos 2.0344 + \mathrm{i}\sin 2.0344)$$
$$z^2 = r^2(\cos\theta + \mathrm{i}\sin\theta)^2 = r^2(\cos 2\theta + \mathrm{i}\sin 2\theta)$$
$$r^2 = 2\sqrt{5} \Rightarrow r = \sqrt{2\sqrt{5}}$$

Since $2\theta = 2.0344 + 2k\pi$ for integer k:

$$\theta = \frac{2.0344}{2} + \frac{2k\pi}{2}$$

$$k = 0: \quad z_0 = \sqrt{2\sqrt{5}}(\cos 1.017 + \mathrm{i}\sin 1.017) = 1.11 + 1.80\mathrm{i}$$
$$k = 1: \quad z_1 = \sqrt{2\sqrt{5}}(\cos 4.159 + \mathrm{i}\sin 4.159) = -1.11 + 1.80\mathrm{i}$$

$$\therefore z = 1.11 + 1.80\mathrm{i} \quad \text{and} \quad z = -1.11 + 1.80\mathrm{i}.$$

5 c)

$$z^4 = 4 - 4\mathrm{i}$$

Using modulus-argument form:

$$4 - 4\mathrm{i} = 4\sqrt{2}\left(\cos(-\tfrac{\pi}{4}) + \mathrm{i}\sin(-\tfrac{\pi}{4})\right)$$
$$z^4 = r^4(\cos\theta + \mathrm{i}\sin\theta)^4 = r^4(\cos 4\theta + \mathrm{i}\sin 4\theta)$$
$$r^4 = 4\sqrt{2} \Rightarrow r = 2^{5/8}$$

Since $4\theta = -\frac{\pi}{4} + 2k\pi$ for integer k:

$$\theta = -\frac{\pi}{16} + \frac{2k\pi}{4}$$

$$k = 0: \quad z_0 = 2^{5/8}\mathrm{e}^{-\pi\mathrm{i}/16} = 1.51 - 0.301\mathrm{i}$$
$$k = 1: \quad z_1 = 2^{5/8}\mathrm{e}^{7\pi\mathrm{i}/16} = 0.301 + 1.51\mathrm{i}$$
$$k = 2: \quad z_2 = 2^{5/8}\mathrm{e}^{15\pi\mathrm{i}/16} = -1.51 + 0.301\mathrm{i}$$
$$k = 3: \quad z_3 = 2^{5/8}\mathrm{e}^{23\pi\mathrm{i}/16} = 2\mathrm{e}^{-7\pi\mathrm{i}/16} = -0.301 - 1.51\mathrm{i}$$

$$\therefore z = 1.51 - 0.301\mathrm{i}, \quad z = 0.301 + 1.51\mathrm{i}, \quad z = -1.51 + 0.301\mathrm{i} \quad \text{and} \quad z = -0.301 - 1.51\mathrm{i}.$$

5 d)

$$z^3 = 4 + 4\mathrm{i}$$

Using modulus-argument form:

$$4 + 4\mathrm{i} = 4\sqrt{2}(\cos\tfrac{\pi}{4} + \mathrm{i}\sin\tfrac{\pi}{4})$$
$$z^3 = r^3(\cos\theta + \mathrm{i}\sin\theta)^3 = r^3(\cos 3\theta + \mathrm{i}\sin 3\theta)$$
$$r^3 = 4\sqrt{2} \Rightarrow r = 2^{5/6}$$

Since $3\theta = \frac{\pi}{4} + 2k\pi$ for integer k:

$$\theta = \frac{\pi}{12} + \frac{2k\pi}{3}$$

$$k = 0: \quad z_0 = 2^{5/6}\mathrm{e}^{\pi\mathrm{i}/12} = 1.72 + 0.461\mathrm{i}$$
$$k = 1: \quad z_1 = 2^{5/6}\mathrm{e}^{9\pi\mathrm{i}/12} = -1.26 + 1.26\mathrm{i}$$
$$k = 2: \quad z_2 = 2^{5/6}\mathrm{e}^{17\pi\mathrm{i}/12} = 2^{5/6}\mathrm{e}^{-7\pi\mathrm{i}/12} = -0.461 + 1.72\mathrm{i}$$
$$\therefore z = 1.72 + 0.461\mathrm{i}, \quad z = -1.26 + 1.26\mathrm{i} \quad \text{and} \quad z = -0.461 + 1.72\mathrm{i}.$$

6)

Let $w = z - \mathrm{i}$ so that $w^2 = -4 + 12\mathrm{i}$. Using modulus-argument form:

$$-4 + 12\mathrm{i} = 4\sqrt{10}(\cos 1.8925 + \mathrm{i}\sin 1.8925)$$
$$z^2 = r^2(\cos\theta + \mathrm{i}\sin\theta)^2 = r^2(\cos 2\theta + \mathrm{i}\sin 2\theta)$$
$$r^2 = 4\sqrt{10} \Rightarrow r = 2 \times 10^{1/4}$$

Since $2\theta = 1.8925 + 2k\pi$ for integer k:

$$\theta = \frac{1.8925}{2} + \frac{2k\pi}{2}$$

$$k = 0: \quad z_0 = 2 \times 10^{1/4}(\cos 0.946 + \mathrm{i}\sin 0.946) = 2.08 + 2.89\mathrm{i}$$
$$k = 1: \quad z_1 = 2 \times 10^{1/4}(\cos 4.088 + \mathrm{i}\sin 4.088) = -2.08 - 2.89\mathrm{i}$$

Since $w = z - \mathrm{i}$, then $z = w + \mathrm{i}$.

$$\therefore z = 2.08 + 3.89\mathrm{i} \quad \text{and} \quad z = -2.08 - 1.89\mathrm{i}.$$

7)

Let $w = z - 2$ so that $w^3 = 8\mathrm{i}$. Using modulus-argument form:

$$8\mathrm{i} = 8(\cos\tfrac{\pi}{2} + \mathrm{i}\sin\tfrac{\pi}{2})$$
$$z^3 = r^3(\cos\theta + \mathrm{i}\sin\theta)^3 = r^3(\cos 3\theta + \mathrm{i}\sin 3\theta)$$
$$r^3 = 8 \Rightarrow r = 2$$

Since $3\theta = \frac{\pi}{2} + 2k\pi$ for integer k:

$$\theta = \frac{\pi}{6} + \frac{2k\pi}{3}$$

$$k = 0: \quad z_0 = 2(\cos\tfrac{\pi}{6} + \mathrm{i}\sin\tfrac{\pi}{6}) = \sqrt{3} + \mathrm{i}$$
$$k = 1: \quad z_1 = 2(\cos\tfrac{5\pi}{6} + \mathrm{i}\sin\tfrac{5\pi}{6}) = -\sqrt{3} + \mathrm{i}$$
$$k = 2: \quad z_2 = 2(\cos\tfrac{9\pi}{6} + \mathrm{i}\sin\tfrac{9\pi}{6}) = -2\mathrm{i}$$

Since $w = z - 2$, then $z = w + 2$.

$$\therefore z = 2 + \sqrt{3} + \mathrm{i}, \quad z = 2 - \sqrt{3} + \mathrm{i} \quad \text{and} \quad z = 2 - 2\mathrm{i}.$$

8 a)

$$\begin{aligned}(-3\sqrt{3}+3\mathrm{i})^3 &= 1(-3\sqrt{3})^3 + 3(-3\sqrt{3})^2(3\mathrm{i}) + 3(-3\sqrt{3})(3\mathrm{i})^2 + (3\mathrm{i})^3 \\ &= 216\mathrm{i}\end{aligned}$$

$$\therefore a = 0 \quad \text{and} \quad b = 216.$$

8 b)

$$z^3 = 216\mathrm{i}$$

Using modulus-argument form:

$$216\mathrm{i} = 216(\cos\tfrac{\pi}{2} + \mathrm{i}\sin\tfrac{\pi}{2})$$

$$z^3 = r^3(\cos\theta + \mathrm{i}\sin\theta)^3 = r^3(\cos 3\theta + \mathrm{i}\sin 3\theta)$$

$$r^3 = 216 \Rightarrow r = 6$$

Since $3\theta = \frac{\pi}{2} + 2k\pi$ for integer k:

$$\theta = \frac{\pi}{6} + \frac{2k\pi}{3}$$

$$\begin{aligned}k = 0: &\quad z_0 = 6\mathrm{e}^{\pi\mathrm{i}/6} = 3\sqrt{3} + 3\mathrm{i} \\ k = 1: &\quad z_1 = 6\mathrm{e}^{5\pi\mathrm{i}/6} = -3\sqrt{3} + 3\mathrm{i} \\ k = 2: &\quad z_2 = 6\mathrm{e}^{9\pi\mathrm{i}/6} = 6\mathrm{e}^{-\pi\mathrm{i}/2} = -6\mathrm{i}\end{aligned}$$

$$\therefore z = 3\sqrt{3} + 3\mathrm{i}, \quad z = -3\sqrt{3} + 3\mathrm{i} \quad \text{and} \quad z = -6\mathrm{i}.$$

8 c)

$z^3 = 216\mathrm{i}$ is not a polynomial with real coefficients. Complex roots are in conjugate pairs only for polynomials with real coefficients.

9 a)

$$\begin{aligned}(1+\mathrm{i})^4 &= 1(1)^4 + 4(1)^3(\mathrm{i}) + 6(1)^2(\mathrm{i})^2 + 4(1)(\mathrm{i})^3 + (\mathrm{i})^4 \\ &= -4\end{aligned}$$

$$\therefore a = -4 \quad \text{and} \quad b = 0.$$

9 b)

$$z^4 = -4$$

Using modulus-argument form:

$$-4 = 4(\cos\pi + \mathrm{i}\sin\pi)$$

$$z^4 = r^4(\cos\theta + \mathrm{i}\sin\theta)^4 = r^4(\cos 4\theta + \mathrm{i}\sin 4\theta)$$

$$r^4 = 4 \Rightarrow r = \sqrt{2}$$

Since $4\theta = \pi + 2k\pi$ for integer k:

$$\theta = \frac{\pi}{4} + \frac{2k\pi}{4}$$

$$\begin{aligned}
k = 0: &\quad z_0 = \sqrt{2}(\cos\tfrac{\pi}{4} + \mathrm{i}\sin\tfrac{\pi}{4}) = 1 + \mathrm{i} \\
k = 1: &\quad z_1 = \sqrt{2}(\cos\tfrac{3\pi}{4} + \mathrm{i}\sin\tfrac{3\pi}{4}) = -1 + \mathrm{i} \\
k = 2: &\quad z_2 = \sqrt{2}(\cos\tfrac{5\pi}{4} + \mathrm{i}\sin\tfrac{5\pi}{4}) = \sqrt{2}(\cos(-\tfrac{3\pi}{4}) + \mathrm{i}\sin(-\tfrac{3\pi}{4})) = -1 - \mathrm{i} \\
k = 3: &\quad z_3 = \sqrt{2}(\cos\tfrac{7\pi}{4} + \mathrm{i}\sin\tfrac{7\pi}{4}) = \sqrt{2}(\cos(-\tfrac{7\pi}{4}) + \mathrm{i}\sin(-\tfrac{7\pi}{4})) = 1 - \mathrm{i}
\end{aligned}$$

$$\therefore z = 1 + \mathrm{i}, \quad z = -1 + \mathrm{i}, \quad z = -1 - \mathrm{i} \quad \text{and} \quad z = 1 - \mathrm{i}.$$

9 c)

$z^4 = -4$ is a polynomial with real coefficients, so complex roots come in conjugate pairs.

10 a)

$$z^3 = 5 + 5\sqrt{3}\mathrm{i}$$

Using modulus-argument form:

$$5 + 5\sqrt{3}\mathrm{i} = 10(\cos\tfrac{\pi}{3} + \mathrm{i}\sin\tfrac{\pi}{3})$$

$$z^3 = r^3(\cos\theta + \mathrm{i}\sin\theta)^3 = r^3(\cos 3\theta + \mathrm{i}\sin 3\theta)$$

$$r^3 = 10 \Rightarrow r = 10^{1/3}$$

Since $3\theta = \frac{\pi}{3} + 2k\pi$ for integer k:

$$\theta = \frac{\pi}{9} + \frac{2k\pi}{3}$$

$$\begin{aligned}
k = 0: &\quad z_0 = 10^{1/3}\mathrm{e}^{\pi\mathrm{i}/9} \\
k = 1: &\quad z_1 = 10^{1/3}\mathrm{e}^{7\pi\mathrm{i}/9} \\
k = 2: &\quad z_2 = 10^{1/3}\mathrm{e}^{13\pi\mathrm{i}/9} = 10^{1/3}\mathrm{e}^{-5\pi\mathrm{i}/9}
\end{aligned}$$

$$\therefore z = 10^{1/3}\mathrm{e}^{\pi\mathrm{i}/9}, \quad z = 10^{1/3}\mathrm{e}^{7\pi\mathrm{i}/9} \quad \text{and} \quad z = 10^{1/3}\mathrm{e}^{-5\pi\mathrm{i}/9}.$$

10 b)

$$\begin{aligned}
(10^{1/3}\mathrm{e}^{\pi\mathrm{i}/9})^9 &= 10^3\mathrm{e}^{\pi\mathrm{i}} = -1000 \\
(10^{1/3}\mathrm{e}^{7\pi\mathrm{i}/9})^9 &= 10^3\mathrm{e}^{7\pi\mathrm{i}} = 10^3\mathrm{e}^{\pi\mathrm{i}} = -1000 \\
(10^{1/3}\mathrm{e}^{-5\pi\mathrm{i}/9})^9 &= 10^3\mathrm{e}^{-5\pi\mathrm{i}} = 10^3\mathrm{e}^{\pi\mathrm{i}} = -1000
\end{aligned}$$

$$\therefore \text{ all the roots satisfy } w^9 + 1000 = 0.$$

11)

$$z^8 = (2+2\mathrm{i})^4$$

But

$$\begin{aligned}(2+2\mathrm{i})^4 &= 1(2)^4 + 4(2)^3(2\mathrm{i}) + 6(2)^2(2\mathrm{i})^2 + 4(2)^1(2\mathrm{i})^3 + (2\mathrm{i})^4 \\ &= 1(2)^4 + 4(2)^3(2\mathrm{i}) + 6(2)^2(2\mathrm{i})^2 + 4(2)^1(2\mathrm{i})^3 + (2\mathrm{i})^4 \\ &= 16 + 64\mathrm{i} - 96 - 64\mathrm{i} + 16 \\ &= -64.\end{aligned}$$

Thus $z^8 = -64$. Using modulus-argument form:

$$-64 = 64(\cos\pi + \mathrm{i}\sin\pi)$$

$$z^8 = r^8(\cos\theta + \mathrm{i}\sin\theta)^8 = r^8(\cos 8\theta + \mathrm{i}\sin 8\theta)$$

$$r^8 = 64 \Rightarrow r = 2^{3/4}$$

Since $8\theta = \pi + 2k\pi$ for integer k:

$$\theta = \frac{\pi}{8} + \frac{2k\pi}{8}$$

$$\begin{aligned}k = 0 &: z_0 = 2^{3/4}\mathrm{e}^{\pi\mathrm{i}/8} \\ k = 1 &: z_1 = 2^{3/4}\mathrm{e}^{3\pi\mathrm{i}/8} \\ k = 2 &: z_2 = 2^{3/4}\mathrm{e}^{5\pi\mathrm{i}/8} \\ k = 3 &: z_3 = 2^{3/4}\mathrm{e}^{7\pi\mathrm{i}/8} \\ k = 4 &: z_4 = 2^{3/4}\mathrm{e}^{-7\pi\mathrm{i}/8} \\ k = 5 &: z_5 = 2^{3/4}\mathrm{e}^{11\pi\mathrm{i}/8} = 2^{3/4}\mathrm{e}^{-5\pi\mathrm{i}/8} \\ k = 6 &: z_6 = 2^{3/4}\mathrm{e}^{13\pi\mathrm{i}/8} = 2^{3/4}\mathrm{e}^{-3\pi\mathrm{i}/8} \\ k = 7 &: z_7 = 2^{3/4}\mathrm{e}^{15\pi\mathrm{i}/8} = 2^{3/4}\mathrm{e}^{-\pi\mathrm{i}/8}\end{aligned}$$

$$\therefore z = 2^{3/4}\mathrm{e}^{\pi\mathrm{i}/8},\ z = 2^{3/4}\mathrm{e}^{3\pi\mathrm{i}/8},\ z = 2^{3/4}\mathrm{e}^{5\pi\mathrm{i}/8},\ z = 2^{3/4}\mathrm{e}^{7\pi\mathrm{i}/8},\ z = 2^{3/4}\mathrm{e}^{-7\pi\mathrm{i}/8},\ z = 2^{3/4}\mathrm{e}^{-5\pi\mathrm{i}/8},$$
$$z = 2^{3/4}\mathrm{e}^{-3\pi\mathrm{i}/8} \quad \text{and} \quad z = 2^{3/4}\mathrm{e}^{-\pi\mathrm{i}/8}.$$

12 a)

$$1 + \mathrm{i} = \sqrt{2}(\cos\tfrac{\pi}{4} + \mathrm{i}\sin\tfrac{\pi}{4}) = \sqrt{2}\mathrm{e}^{\pi\mathrm{i}/4}.$$

12 b)

$$\sqrt{3} - \mathrm{i} = 2(\cos(-\tfrac{\pi}{6}) + \mathrm{i}\sin(-\tfrac{\pi}{6})) = 2\mathrm{e}^{-\pi\mathrm{i}/6}.$$

12 c)

$$z^4 = (1+\mathrm{i})^2(\sqrt{3}-\mathrm{i})^2 = (\sqrt{2}\mathrm{e}^{\pi\mathrm{i}/4})^2(2\mathrm{e}^{-\pi\mathrm{i}/6})^2$$
$$= 2\mathrm{e}^{\pi\mathrm{i}/2} \times 4\mathrm{e}^{-\pi\mathrm{i}/3} = 8\mathrm{e}^{\pi\mathrm{i}/6}.$$

$$z^4 = r^4(\cos\theta + \mathrm{i}\sin\theta)^4 = r^4(\cos 4\theta + \mathrm{i}\sin 4\theta)$$
$$r^4 = 8 \Rightarrow r = 8^{1/4} = 2^{3/4}$$

Since $4\theta = \frac{\pi}{6} + 2k\pi$ for integer k:

$$\theta = \frac{\pi}{24} + \frac{2k\pi}{4}$$

$$k=0: \quad z_0 = 2^{3/4}\mathrm{e}^{\pi\mathrm{i}/24}$$
$$k=1: \quad z_1 = 2^{3/4}\mathrm{e}^{13\pi\mathrm{i}/24}$$
$$k=2: \quad z_2 = 2^{3/4}\mathrm{e}^{25\pi\mathrm{i}/24} = 2^{3/4}\mathrm{e}^{-23\pi\mathrm{i}/24}$$
$$k=3: \quad z_3 = 2^{3/4}\mathrm{e}^{37\pi\mathrm{i}/24} = 2^{3/4}\mathrm{e}^{-11\pi\mathrm{i}/24}$$

$$\therefore z = 2^{3/4}\mathrm{e}^{\pi\mathrm{i}/24},\ z = 2^{3/4}\mathrm{e}^{13\pi\mathrm{i}/24},\ z = 2^{3/4}\mathrm{e}^{-23\pi\mathrm{i}/24} \quad \text{and} \quad z = 2^{3/4}\mathrm{e}^{-11\pi\mathrm{i}/24}.$$

13 a)

$$5\sqrt{3} + 5\mathrm{i} = 10(\cos\tfrac{\pi}{6} + \mathrm{i}\sin\tfrac{\pi}{6}).$$

13 b)

$$z^5 = 10^5(\cos\tfrac{5\pi}{6} + \mathrm{i}\sin\tfrac{5\pi}{6})$$
$$= 100000(\cos\tfrac{5\pi}{6} + \mathrm{i}\sin\tfrac{5\pi}{6})$$

$$w^4 = 100000(\cos\tfrac{5\pi}{6} + \mathrm{i}\sin\tfrac{5\pi}{6})$$
$$w^4 = r^4(\cos\theta + \mathrm{i}\sin\theta)^4 = r^4(\cos 4\theta + \mathrm{i}\sin 4\theta)$$
$$r^4 = 100000 \Rightarrow r = 10^{5/4}$$

Since $4\theta = \frac{5\pi}{6} + 2k\pi$ for integer k:

$$\theta = \frac{5\pi}{24} + \frac{2k\pi}{4}$$

$$k=0: \quad z_0 = 10^{5/4}(\cos\tfrac{5\pi}{24} + \mathrm{i}\sin\tfrac{5\pi}{24})$$
$$k=1: \quad z_1 = 10^{5/4}(\cos\tfrac{17\pi}{24} + \mathrm{i}\sin\tfrac{17\pi}{24})$$
$$k=2: \quad z_2 = 10^{5/4}(\cos\tfrac{29\pi}{24} + \mathrm{i}\sin\tfrac{29\pi}{24}) = 10^{5/4}(\cos(-\tfrac{19\pi}{24}) + \mathrm{i}\sin(-\tfrac{19\pi}{24}))$$
$$k=3: \quad z_3 = 10^{5/4}(\cos\tfrac{41\pi}{24} + \mathrm{i}\sin\tfrac{41\pi}{24}) = 10^{5/4}(\cos(-\tfrac{7\pi}{24}) + \mathrm{i}\sin(-\tfrac{7\pi}{24}))$$

$$\therefore z = 10^{5/4}(\cos\tfrac{5\pi}{24} + \mathrm{i}\sin\tfrac{5\pi}{24}),\ z = 10^{5/4}(\cos\tfrac{17\pi}{24} + \mathrm{i}\sin\tfrac{17\pi}{24}),$$
$$z = 10^{5/4}(\cos(-\tfrac{19\pi}{24}) + \mathrm{i}\sin(-\tfrac{19\pi}{24})) \quad \text{and} \quad z = 10^{5/4}(\cos(-\tfrac{7\pi}{24}) + \mathrm{i}\sin(-\tfrac{7\pi}{24})).$$

14)

$$z^{5/3} = -1 - \sqrt{3}\mathrm{i}$$

$$\begin{aligned} z^5 &= (-1-\sqrt{3}\mathrm{i})^3 \\ &= 1(-1)^3 + 3(-1)^2(-\sqrt{3}\mathrm{i}) + 3(-1)(-\sqrt{3}\mathrm{i})^2 + 1(-\sqrt{3}\mathrm{i})^3 \\ &= -1 - 3\sqrt{3}\mathrm{i} + 9 + 3\sqrt{3}\mathrm{i} \\ &= 8 \end{aligned}$$

$$\therefore z^5 = 8.$$

Using modulus-argument form:

$$8 = 8(\cos 0 + \mathrm{i}\sin 0)$$

$$z^5 = r^5(\cos\theta + \mathrm{i}\sin\theta)^5 = r^5(\cos 5\theta + \mathrm{i}\sin 5\theta)$$

$$r^5 = 8 \Rightarrow r = 8^{1/5} = 2^{3/5}$$

Since $5\theta = 0 + 2k\pi$ for integer k:

$$\theta = 0 + \frac{2k\pi}{5}$$

$$\begin{aligned} k = 0: &\quad z_0 = 2^{3/5}\mathrm{e}^{0\mathrm{i}} \\ k = 1: &\quad z_1 = 2^{3/5}\mathrm{e}^{2\pi\mathrm{i}/5} \\ k = 2: &\quad z_2 = 2^{3/5}\mathrm{e}^{4\pi\mathrm{i}/5} \\ k = 3: &\quad z_3 = 2^{3/5}\mathrm{e}^{6\pi\mathrm{i}/5} = 2^{3/5}\mathrm{e}^{-4\pi\mathrm{i}/5} \\ k = 4: &\quad z_4 = 2^{3/5}\mathrm{e}^{8\pi\mathrm{i}/5} = 2^{3/5}\mathrm{e}^{-2\pi\mathrm{i}/5} \end{aligned}$$

$$\therefore z = 2^{3/5}\mathrm{e}^{0\mathrm{i}},\ z = 2^{3/5}\mathrm{e}^{2\pi\mathrm{i}/5},\ z = 2^{3/5}\mathrm{e}^{4\pi\mathrm{i}/5},\ z = 2^{3/5}\mathrm{e}^{-4\pi\mathrm{i}/5} \quad \text{and} \quad z = 2^{3/5}\mathrm{e}^{-2\pi\mathrm{i}/5}.$$

15)

$$z^{4/3} = \sqrt{2} + \sqrt{2}\mathrm{i}$$

$$\begin{aligned} z^4 &= (\sqrt{2}+\sqrt{2}\mathrm{i})^3 \\ &= 1(\sqrt{2})^3 + 3(\sqrt{2})^2(\sqrt{2}\mathrm{i}) + 3(\sqrt{2})(\sqrt{2}\mathrm{i})^2 + 1(\sqrt{2}\mathrm{i})^3 \\ &= 2\sqrt{2} + 6\sqrt{2}\mathrm{i} - 6\sqrt{2} - 2\sqrt{2}\mathrm{i} \\ &= -4\sqrt{2} + 4\sqrt{2}\mathrm{i} \end{aligned}$$

$$\therefore z^5 = -4\sqrt{2} + 4\sqrt{2}\mathrm{i}.$$

Using modulus-argument form:

$$-4\sqrt{2} + 4\sqrt{2}\mathrm{i} = 8(\cos\tfrac{3\pi}{4} + \mathrm{i}\sin\tfrac{3\pi}{4})$$

$$z^4 = r^4(\cos\theta + \mathrm{i}\sin\theta)^4 = r^4(\cos 4\theta + \mathrm{i}\sin 4\theta)$$

$$r^4 = 8 \Rightarrow r = 8^{1/4} = 2^{3/4}$$

Since $4\theta = \frac{3\pi}{4} + 2k\pi$ for integer k:

$$\theta = \frac{3\pi}{16} + \frac{2k\pi}{4}$$

$$
\begin{aligned}
k=0: &\quad z_0 = 2^{3/4}\mathrm{e}^{3\pi\mathrm{i}/16}\\
k=1: &\quad z_1 = 2^{3/4}\mathrm{e}^{11\pi\mathrm{i}/16}\\
k=2: &\quad z_2 = 2^{3/4}\mathrm{e}^{19\pi\mathrm{i}/16} = 2^{3/4}\mathrm{e}^{-13\pi\mathrm{i}/16}\\
k=3: &\quad z_3 = 2^{3/4}\mathrm{e}^{27\pi\mathrm{i}/16} = 2^{3/4}\mathrm{e}^{-5\pi\mathrm{i}/16}
\end{aligned}
$$

$$
\therefore z = 2^{3/4}\mathrm{e}^{3\pi\mathrm{i}/16},\ z = 2^{3/4}\mathrm{e}^{11\pi\mathrm{i}/16},\ z = 2^{3/4}\mathrm{e}^{-13\pi\mathrm{i}/16} \quad \text{and} \quad z = 2^{3/4}\mathrm{e}^{-5\pi\mathrm{i}/16}.
$$

16 a)

$$
z_1 = 2(\cos\tfrac{\pi}{3} + \mathrm{i}\sin\tfrac{\pi}{3}).
$$

16 b)

$$
z_2 = 2(\cos(-\tfrac{\pi}{6}) + \mathrm{i}\sin(-\tfrac{\pi}{6})).
$$

16 c)

$$
\begin{aligned}
z_1^3 z_2^2 &= 2^3(\cos\tfrac{3\pi}{3} + \mathrm{i}\sin\tfrac{3\pi}{3}) \times 2^2(\cos(-\tfrac{2\pi}{6}) + \mathrm{i}\sin(-\tfrac{2\pi}{6}))\\
&= 8 \times 4(\cos(\pi - \tfrac{\pi}{3}) + \mathrm{i}\sin(\pi - \tfrac{\pi}{3}))\\
&= 32(\cos\tfrac{2\pi}{3} + \mathrm{i}\sin\tfrac{2\pi}{3})
\end{aligned}
$$

$$
w^6 = 32(\cos\tfrac{2\pi}{3} + \mathrm{i}\sin\tfrac{2\pi}{3})
$$

$$
w^6 = r^6(\cos\theta + \mathrm{i}\sin\theta)^6 = r^6(\cos 6\theta + \mathrm{i}\sin 6\theta)
$$

$$
r^6 = 32 \Rightarrow r = 2^{5/6}
$$

Since $6\theta = \frac{2\pi}{3} + 2k\pi$ for integer k:

$$
\theta = \frac{2\pi}{18} + \frac{2k\pi}{6}
$$

$$
\begin{aligned}
k=0: &\quad z_0 = 2^{5/6}(\cos\tfrac{\pi}{9} + \mathrm{i}\sin\tfrac{\pi}{9})\\
k=1: &\quad z_1 = 2^{5/6}(\cos\tfrac{4\pi}{9} + \mathrm{i}\sin\tfrac{4\pi}{9})\\
k=2: &\quad z_2 = 2^{5/6}(\cos\tfrac{7\pi}{9} + \mathrm{i}\sin\tfrac{7\pi}{9})\\
k=3: &\quad z_3 = 2^{5/6}(\cos\tfrac{10\pi}{9} + \mathrm{i}\sin\tfrac{10\pi}{9}) = 2^{5/6}(\cos(-\tfrac{8\pi}{9}) + \mathrm{i}\sin(-\tfrac{8\pi}{9}))\\
k=4: &\quad z_4 = 2^{5/6}(\cos\tfrac{13\pi}{9} + \mathrm{i}\sin\tfrac{13\pi}{9}) = 2^{5/6}(\cos(-\tfrac{5\pi}{9}) + \mathrm{i}\sin(-\tfrac{5\pi}{9}))\\
k=5: &\quad z_5 = 2^{5/6}(\cos\tfrac{16\pi}{9} + \mathrm{i}\sin\tfrac{16\pi}{9}) = 2^{5/6}(\cos(-\tfrac{2\pi}{9}) + \mathrm{i}\sin(-\tfrac{2\pi}{9}))
\end{aligned}
$$

$$
\begin{aligned}
\therefore\ & w = 2^{5/6}(\cos\tfrac{\pi}{9} + \mathrm{i}\sin\tfrac{\pi}{9}),\ w = 2^{5/6}(\cos\tfrac{4\pi}{9} + \mathrm{i}\sin\tfrac{4\pi}{9}),\\
& w = 2^{5/6}(\cos\tfrac{7\pi}{9} + \mathrm{i}\sin\tfrac{7\pi}{9}),\ w = 2^{5/6}(\cos(-\tfrac{8\pi}{9}) + \mathrm{i}\sin(-\tfrac{8\pi}{9})),\\
& w = 2^{5/6}(\cos(-\tfrac{5\pi}{9}) + \mathrm{i}\sin(-\tfrac{5\pi}{9})) \quad \text{and} \quad w = 2^{5/6}(\cos(-\tfrac{2\pi}{9}) + \mathrm{i}\sin(-\tfrac{2\pi}{9})).
\end{aligned}
$$

17 a)

$$(a-1)^3 = a^3 - 3a^2 + 3a - 1.$$

17 b)

$$\begin{aligned} z^3 - 3z^2 + 3z - 9 &= z^3 - 3z^2 + 3z - 1 - 8 \\ &= (z-1)^3 - 8. \end{aligned}$$

Hence solving $(z-1)^3 = 8$. Let $w = z - 1$, so we solve $w^3 = 8$. Using modulus-argument form:

$$8 = 8(\cos 0 + \mathrm{i}\sin 0)$$

$$z^3 = r^3(\cos\theta + \mathrm{i}\sin\theta)^3 = r^3(\cos 3\theta + \mathrm{i}\sin 3\theta)$$

$$r^3 = 8 \Rightarrow r = 2$$

Since $3\theta = 0 + 2k\pi$ for integer k:

$$\theta = 0 + \frac{2k\pi}{3}$$

$$\begin{aligned} k = 0: &\quad z_0 = 2\mathrm{e}^{0\mathrm{i}} = 2 \\ k = 1: &\quad z_1 = 2\mathrm{e}^{2\pi\mathrm{i}/3} = -1 + \sqrt{3}\mathrm{i} \\ k = 2: &\quad z_2 = 2\mathrm{e}^{4\pi\mathrm{i}/3} = 2\mathrm{e}^{-2\pi\mathrm{i}/3} = -1 - \sqrt{3}\mathrm{i} \end{aligned}$$

Since $w = z - 1$, then $z = w + 1$.

$$\therefore z = 3, \quad z = \sqrt{3}\mathrm{i} \quad \text{and} \quad z = -\sqrt{3}\mathrm{i}.$$

18)

First spot that

$$(a-1)^6 = a^6 - 6a^5 + 15a^4 - 20a^3 + 15a^2 - 6a + 1.$$

And so

$$(z-1)^6 + 1 = 0 \Rightarrow (z-1)^6 = -1.$$

Let $w = z - 1$, so we solve $w^6 = -1$. Using modulus-argument form:

$$-1 = 1(\cos\pi + \mathrm{i}\sin\pi)$$

$$z^6 = r^6(\cos\theta + \mathrm{i}\sin\theta)^6 = r^6(\cos 6\theta + \mathrm{i}\sin 6\theta)$$

$$r^6 = 1 \Rightarrow r = 1$$

Since $6\theta = \pi + 2k\pi$ for integer k:

$$\theta = \frac{\pi}{6} + \frac{2k\pi}{6}$$

$$\begin{aligned} k = 0: &\quad z_0 = \mathrm{e}^{\pi\mathrm{i}/6} = \frac{\sqrt{3}}{2} + \frac{1}{2}\mathrm{i} \\ k = 1: &\quad z_1 = \mathrm{e}^{3\pi\mathrm{i}/6} = \mathrm{i} \\ k = 2: &\quad z_2 = \mathrm{e}^{5\pi\mathrm{i}/6} = -\frac{\sqrt{3}}{2} + \frac{1}{2}\mathrm{i} \end{aligned}$$

$$k=3: \quad z_3 = \mathrm{e}^{7\pi\mathrm{i}/6} = -\frac{\sqrt{3}}{2} - \frac{1}{2}\mathrm{i}$$
$$k=4: \quad z_4 = \mathrm{e}^{9\pi\mathrm{i}/6} = -\mathrm{i}$$
$$k=5: \quad z_5 = \mathrm{e}^{11\pi\mathrm{i}/6} = \frac{\sqrt{3}}{2} - \frac{1}{2}\mathrm{i}$$

Since $w = z - 1$, then $z = w + 1$.

$$\therefore z = 1 + \frac{\sqrt{3}}{2} + \frac{1}{2}\mathrm{i}, \quad z = 1 + \mathrm{i}, \quad z = 1 - \frac{\sqrt{3}}{2} + \frac{1}{2}\mathrm{i},$$
$$z = 1 - \frac{\sqrt{3}}{2} - \frac{1}{2}\mathrm{i}, \quad z = 1 - \mathrm{i} \quad \text{and} \quad z = 1 + \frac{\sqrt{3}}{2} - \frac{1}{2}\mathrm{i}.$$

19 a)

$$z = 1 + \mathrm{i} = \sqrt{2}(\cos\tfrac{\pi}{4} + \mathrm{i}\sin\tfrac{\pi}{4}).$$

So

$$z^n = (\sqrt{2}(\cos\tfrac{\pi}{4} + \mathrm{i}\sin\tfrac{\pi}{4}))^n = (\sqrt{2})^n(\cos\tfrac{n\pi}{4} + \mathrm{i}\sin\tfrac{n\pi}{4}).$$

Therefore we need $\sin\frac{n\pi}{4} = 0$, so $\frac{n\pi}{4} = k\pi$ for some integer k and $n = 4k$. But also $\cos\frac{n\pi}{4} > 0$, so n cannot be odd.

$$\therefore n = 8.$$

19 b)

$$z^8 = (\sqrt{2})^8(\cos\tfrac{8\pi}{4} + \mathrm{i}\sin\tfrac{8\pi}{4}) = 16.$$

19 c)

Since $8\theta = 0 + 2k\pi$ for integer k, $\theta = \frac{2k\pi}{8}$.

$$\sqrt{2}(\cos 0 + \mathrm{i}\sin 0) = \sqrt{2}$$
$$\sqrt{2}(\cos\tfrac{\pi}{4} + \mathrm{i}\sin\tfrac{\pi}{4}) = 1 + \mathrm{i}$$
$$\sqrt{2}(\cos\tfrac{2\pi}{4} + \mathrm{i}\sin\tfrac{2\pi}{4}) = \sqrt{2}\mathrm{i}$$
$$\sqrt{2}(\cos\tfrac{3\pi}{4} + \mathrm{i}\sin\tfrac{3\pi}{4}) = -1 + \mathrm{i}$$
$$\sqrt{2}(\cos\tfrac{4\pi}{4} + \mathrm{i}\sin\tfrac{4\pi}{4}) = -\sqrt{2}$$
$$\sqrt{2}(\cos\tfrac{5\pi}{4} + \mathrm{i}\sin\tfrac{5\pi}{4}) = -1 - \mathrm{i}$$
$$\sqrt{2}(\cos\tfrac{6\pi}{4} + \mathrm{i}\sin\tfrac{6\pi}{4}) = -\sqrt{2}\mathrm{i}$$
$$\sqrt{2}(\cos\tfrac{7\pi}{4} + \mathrm{i}\sin\tfrac{7\pi}{4}) = 1 - \mathrm{i}$$

$$z = \sqrt{2},\ z = 1 + \mathrm{i},\ z = \sqrt{2}\mathrm{i},\ z = -1 + \mathrm{i},\ z = -\sqrt{2},\ z = -1 - \mathrm{i},\ z = -\sqrt{2}\mathrm{i} \quad \text{and} \quad z = 1 - \mathrm{i}.$$

20 a)

$$z = -7 + 7\sqrt{3}\mathrm{i} = 14(\cos\tfrac{2\pi}{3} + \mathrm{i}\sin\tfrac{2\pi}{3}).$$

So that

$$z^n = (14(\cos\tfrac{\pi}{3} + \mathrm{i}\sin\tfrac{\pi}{3}))^n = 14^n(\cos\tfrac{2n\pi}{3} + \mathrm{i}\sin\tfrac{2n\pi}{3}).$$

Therefore we need $\sin\frac{2n\pi}{3} = 0$, so $\frac{2n\pi}{3} = k\pi$ for some integer k and $n = \frac{3}{2}k$. But also $\cos\frac{n\pi}{3} > 0$, so n cannot be odd, hence $k = 2$.

$$\therefore n = 3.$$

20 b)

$$z^3 = 14^3(\cos 2\pi + \mathrm{i}\sin 2\pi) = 2744.$$

20 c)

$$\begin{aligned} 14(\cos\tfrac{2\pi}{3} + \mathrm{i}\sin\tfrac{2\pi}{3}) &= -7 + 7\sqrt{3}\mathrm{i} \\ 14(\cos\tfrac{4\pi}{3} + \mathrm{i}\sin\tfrac{4\pi}{3}) &= -7 - 7\sqrt{3}\mathrm{i} \\ \therefore z = 14, \quad z = -7 + 7\sqrt{3}\mathrm{i} \quad &\text{and} \quad z = -7 - 7\sqrt{3}\mathrm{i}. \end{aligned}$$

21 a)

$$\begin{aligned} z &= \frac{2\sqrt{3} + 2\mathrm{i}}{\sqrt{3} - \mathrm{i}} \\ &= \frac{4(\cos\frac{\pi}{6} + \mathrm{i}\sin\frac{\pi}{6})}{2(\cos(-\frac{\pi}{6}) + \mathrm{i}\sin(-\frac{\pi}{6}))} \\ &= 2(\cos\tfrac{\pi}{3} + \mathrm{i}\sin\tfrac{\pi}{3}). \end{aligned}$$

So

$$z^n = (2(\cos\tfrac{\pi}{3} + \mathrm{i}\sin\tfrac{\pi}{3}))^n = 2^n(\cos\tfrac{n\pi}{3} + \mathrm{i}\sin\tfrac{n\pi}{3}).$$

Therefore we need $\sin\frac{n\pi}{3} = 0$, so $\frac{n\pi}{3} = k\pi$ for some integer k and $n = 3k$. But also $\cos\frac{n\pi}{3} > 0$, so n cannot be odd, hence $k = 2$.

$$\therefore n = 6.$$

21 b)

$$z^6 = 2^6(\cos\tfrac{6\pi}{3} + \mathrm{i}\sin\tfrac{6\pi}{3}) = 64.$$

21 c)

$$\begin{aligned} 2(\cos 0 + \mathrm{i}\sin 0) &= 2 \\ 2(\cos\tfrac{\pi}{3} + \mathrm{i}\sin\tfrac{\pi}{3}) &= 1 + \sqrt{3}\mathrm{i} \\ 2(\cos\tfrac{2\pi}{3} + \mathrm{i}\sin\tfrac{2\pi}{3}) &= -1 + \sqrt{3}\mathrm{i} \\ 2(\cos\pi + \mathrm{i}\sin\pi) &= -2 \\ 2(\cos\tfrac{4\pi}{3} + \mathrm{i}\sin\tfrac{4\pi}{3}) &= -1 - \sqrt{3}\mathrm{i} \\ 2(\cos\tfrac{5\pi}{3} + \mathrm{i}\sin\tfrac{5\pi}{3}) &= 1 - \sqrt{3}\mathrm{i} \end{aligned}$$

$$\therefore z = 2,\ z = 1 + \sqrt{3}\mathrm{i},\ z = -1 + \sqrt{3}\mathrm{i},\ z = -2,\ z = -1 - \sqrt{3}\mathrm{i} \quad \text{and} \quad z = 1 - \sqrt{3}\mathrm{i}.$$

22)

$$16z^4 = (z-\mathrm{i})^4 \Rightarrow \frac{z^4}{(z-\mathrm{i})^4} = 16 \Rightarrow \left(\frac{z-\mathrm{i}}{z}\right)^4 = 16.$$

Let $w = \frac{z-\mathrm{i}}{z}$, so $w^4 = 16$. Using modulus-argument form:

$$16 = 16(\cos 0 + \mathrm{i}\sin 0)$$
$$w^4 = r^4(\cos\theta + \mathrm{i}\sin\theta)^4 = r^4(\cos 4\theta + \mathrm{i}\sin 4\theta)$$
$$r^4 = 16 \Rightarrow r = 2$$

Since $4\theta = 0 + 2k\pi$ for integer k:

$$\theta = 0 + \frac{2k\pi}{4}.$$

So $w = 2\mathrm{e}^{k\pi\mathrm{i}/2}$ for $k = 0, 1, 2, 3$. Substituting back in:

$$2\mathrm{e}^{k\pi\mathrm{i}/2}z = z - \mathrm{i}$$
$$2\mathrm{e}^{k\pi\mathrm{i}/2}z - z = -\mathrm{i} \Rightarrow (2\mathrm{e}^{k\pi\mathrm{i}/2} - 1)z = -\mathrm{i}$$
$$\therefore z = -\frac{\mathrm{i}}{2\mathrm{e}^{k\pi\mathrm{i}/2} - 1} = \frac{\mathrm{i}}{1 - 2\mathrm{e}^{k\pi\mathrm{i}/2}}.$$

$$k = 0,\ e^0 = 1: \quad z_0 = \frac{\mathrm{i}}{1-2} = -\mathrm{i}$$
$$k = 1,\ e^{\pi\mathrm{i}/2} = \mathrm{i}: \quad z_1 = \frac{\mathrm{i}}{1-2\mathrm{i}} \times \frac{1+2\mathrm{i}}{1+2\mathrm{i}} = \frac{\mathrm{i}-2}{1+4} = -\frac{2}{5} + \frac{1}{5}\mathrm{i}$$
$$k = 2,\ e^{\pi\mathrm{i}} = -1: \quad z_2 = \frac{\mathrm{i}}{1+2} = \frac{1}{3}\mathrm{i}$$
$$k = 3,\ e^{3\pi\mathrm{i}/2} = -\mathrm{i}: \quad z_3 = \frac{\mathrm{i}}{1+2\mathrm{i}} \times \frac{1-2\mathrm{i}}{1-2\mathrm{i}} = \frac{\mathrm{i}+2}{1+4} = \frac{2}{5} + \frac{1}{5}\mathrm{i}$$

So the solutions are

$$z = -\mathrm{i}, \quad z = -\frac{2}{5} + \frac{1}{5}\mathrm{i}, \quad z = \frac{1}{3}\mathrm{i} \quad \text{and} \quad z = \frac{2}{5} + \frac{1}{5}\mathrm{i}.$$

23 a)

$$z^3 = 1 + \sqrt{3}\mathrm{i}$$

Using modulus-argument form:

$$1 + \sqrt{3}\mathrm{i} = 2(\cos\tfrac{\pi}{3} + \mathrm{i}\sin\tfrac{\pi}{3})$$
$$z^3 = r^3(\cos\theta + \mathrm{i}\sin\theta)^3 = r^3(\cos 3\theta + \mathrm{i}\sin 3\theta)$$
$$r^3 = 2 \Rightarrow r = 2^{1/3}$$

Since $3\theta = \frac{\pi}{3} + 2k\pi$ for integer k:

$$\theta = \frac{\pi}{9} + \frac{2k\pi}{3}$$

$$k = 0: \quad z_0 = 2^{1/3}\mathrm{e}^{\pi\mathrm{i}/9}$$
$$k = 1: \quad z_1 = 2^{1/3}\mathrm{e}^{7\pi\mathrm{i}/9}$$
$$k = 2: \quad z_2 = 2^{1/3}\mathrm{e}^{13\pi\mathrm{i}/9} = 2^{1/3}\mathrm{e}^{-5\pi\mathrm{i}/9}$$
$$\therefore z = 2^{1/3}\mathrm{e}^{\pi\mathrm{i}/9}, \quad z = 2^{1/3}\mathrm{e}^{7\pi\mathrm{i}/9} \quad \text{and} \quad z = 2^{1/3}\mathrm{e}^{-5\pi\mathrm{i}/9}.$$

23 b)

The sum of the three roots is zero:

$$2^{1/3}\mathrm{e}^{\pi\mathrm{i}/9} + 2^{1/3}\mathrm{e}^{7\pi\mathrm{i}/9} + 2^{1/3}\mathrm{e}^{-5\pi\mathrm{i}/9} = 0$$

$$\mathrm{e}^{\pi\mathrm{i}/9} + \mathrm{e}^{7\pi\mathrm{i}/9} + \mathrm{e}^{-5\pi\mathrm{i}/9} = 0.$$

So

$$\mathrm{Re}(\mathrm{e}^{\pi\mathrm{i}/9} + \mathrm{e}^{7\pi\mathrm{i}/9} + \mathrm{e}^{-5\pi\mathrm{i}/9}) = 0$$

$$\cos\frac{\pi}{9} + \cos\frac{7\pi}{9} + \cos\left(-\frac{5\pi}{9}\right) = 0$$

$$\therefore \cos\frac{\pi}{9} + \cos\frac{5\pi}{9} + \cos\frac{7\pi}{9} = 0 \quad \text{as required.}$$

24 a)

$$z^3 = 6\sqrt{3} + 6\mathrm{i} = 6(\sqrt{3} + \mathrm{i})$$

Using modulus-argument form:

$$\sqrt{3} + \mathrm{i} = 2(\cos\tfrac{\pi}{6} + \mathrm{i}\sin\tfrac{\pi}{6})$$

So $6(\sqrt{3} + \mathrm{i}) = 12(\cos\frac{\pi}{6} + \mathrm{i}\sin\frac{\pi}{6})$.

$$z^3 = r^3(\cos\theta + \mathrm{i}\sin\theta)^3 = r^3(\cos 3\theta + \mathrm{i}\sin 3\theta)$$

$$r^3 = 12 \Rightarrow r = 12^{1/3}$$

Since $3\theta = \frac{\pi}{6} + 2k\pi$ for integer k:

$$\theta = \frac{\pi}{18} + \frac{2k\pi}{3}$$

$$\begin{aligned} k = 0: &\quad z_0 = 12^{1/3}\mathrm{e}^{\pi\mathrm{i}/18} \\ k = 1: &\quad z_1 = 12^{1/3}\mathrm{e}^{13\pi\mathrm{i}/18} \\ k = 2: &\quad z_2 = 12^{1/3}\mathrm{e}^{25\pi\mathrm{i}/18} = 12^{1/3}\mathrm{e}^{-11\pi\mathrm{i}/18} \end{aligned}$$

$$\therefore z = 12^{1/3}\mathrm{e}^{\pi\mathrm{i}/18}, \quad z = 12^{1/3}\mathrm{e}^{13\pi\mathrm{i}/18} \quad \text{and} \quad z = 12^{1/3}\mathrm{e}^{-11\pi\mathrm{i}/18}.$$

24 b)

The sum of the three roots is zero:

$$12^{1/3}\mathrm{e}^{\pi\mathrm{i}/18} + 12^{1/3}\mathrm{e}^{13\pi\mathrm{i}/18} + 12^{1/3}\mathrm{e}^{-11\pi\mathrm{i}/18} = 0$$

$$\mathrm{e}^{\pi\mathrm{i}/18} + \mathrm{e}^{13\pi\mathrm{i}/18} + \mathrm{e}^{-11\pi\mathrm{i}/18} = 0.$$

So

$$\mathrm{Im}(\mathrm{e}^{\pi\mathrm{i}/18} + \mathrm{e}^{13\pi\mathrm{i}/18} + \mathrm{e}^{-11\pi\mathrm{i}/18}) = 0$$

$$\sin\frac{\pi}{18} + \sin\frac{13\pi}{18} + \sin\left(-\frac{11\pi}{18}\right) = 0$$

$$\therefore \sin\frac{\pi}{18} - \sin\frac{11\pi}{18} + \sin\frac{13\pi}{18} = 0 \quad \text{as required.}$$

25 a)

$$z_1 = 1 + \sqrt{3}\mathrm{i} = 2\mathrm{e}^{\pi\mathrm{i}/3}.$$

25 b)

$$z_2 = 1 + \sqrt{3}\mathrm{i} = 2\mathrm{e}^{-\pi\mathrm{i}/3}.$$

25 c)

$$\begin{aligned} z_1^6 + z_2^6 &= (2\mathrm{e}^{\pi\mathrm{i}/3})^6 + (2\mathrm{e}^{-\pi\mathrm{i}/3})^6 \\ &= 2^6\mathrm{e}^{6\pi\mathrm{i}/3} + 2^6\mathrm{e}^{-6\pi\mathrm{i}/3} \\ &= 64\mathrm{e}^0 + 64\mathrm{e}^0 = 128 \end{aligned}$$

$$\begin{aligned} \text{and } z_1^6 - z_2^6 &= 2^6\mathrm{e}^{6\pi\mathrm{i}/3} - 2^6\mathrm{e}^{-6\pi\mathrm{i}/3} \\ &= 64\mathrm{e}^0 - 64\mathrm{e}^0 = 0. \end{aligned}$$

25 d)

$$\omega = \mathrm{e}^{2\pi\mathrm{i}/3} = -\frac{1}{2} + \frac{\sqrt{3}}{2}\mathrm{i}, \quad \omega^2 = \mathrm{e}^{4\pi\mathrm{i}/3} = -\frac{1}{2} - \frac{\sqrt{3}}{2}\mathrm{i} \quad \text{and} \quad z_1 z_2 = 4.$$

$$\begin{aligned} z_1^6 + \omega z_2^6 + \omega^2 (z_1 z_2)^6 &= 64 + 64\omega + 4096\omega^2 \\ &= 64 + 64\left(-\frac{1}{2} + \frac{\sqrt{3}}{2}\mathrm{i}\right) + 4096\left(-\frac{1}{2} - \frac{\sqrt{3}}{2}\mathrm{i}\right) \\ &= 64 - 32 + 32\sqrt{3}\mathrm{i} - 2048 - 2048\sqrt{3}\mathrm{i} \\ &= -2016 - 2016\sqrt{3}\mathrm{i}. \end{aligned}$$